

JioMausam, explained simply

Everything about our idea, our prototype and every technical word you'll hear, in plain language.

You don't need any background in science, telecom, statistics or insurance to read this. Every idea is built up from scratch with everyday examples. It takes about 40 minutes to read end to end.

Read it in order the first time. Then keep the dictionary (pages 19–21) and the practice pages (22–24) open while you rehearse.



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The whole project in one minute

If you read only one page, read this one. Everything else in this guide is a deeper explanation of these five sentences.

- 1 Bad weather costs poor workers money.** When heavy rain or extreme heat hits, delivery riders, street vendors and labourers can't work, and they lose that day's income. Nobody pays them back.
- 2 India can't see weather street by street.** Our weather instruments are few and far apart, so nobody knows exactly which streets got hit and which didn't.
- 3 Mobile towers can already "feel" the weather.** Towers talk to each other with invisible radio beams. Rain makes those beams weaker, and the weaker the beam, the harder it's raining. The network already records this.
- 4 We turn that into a live weather map and an automatic insurance product.** The map is **JioMausam**. The insurance is **Mausam Kavach**: pay ₹29 a month with your recharge, and if heavy rain or heat hits where you are, ₹300 lands in your JioPay wallet within hours. No forms.
- 5 We built a working prototype that proves it.** We used real signals from 500 tower-to-tower links and compared our rain readings with an official weather radar. They matched very closely.

Three names to remember

NAME	WHAT IT IS	MEANING OF THE WORD
JioMausam	The whole platform: the system that reads tower signals and draws the live weather map.	<i>Mausam</i> = weather (Hindi)
Mausam Kavach	The insurance product people buy for ₹29 a month.	<i>Kavach</i> = shield or armour
Control Room	Our prototype: the working website we built to show the idea on real data.	Like a TV control room full of live screens

SAY IT LIKE THIS

"Every Jio tower is a rain gauge and every phone is a thermometer. We map India's weather street by street, and pay workers within hours when it stops their work."

Meet Ravi

Ravi is a food-delivery rider in Mumbai. He earns about ₹800–1,000 on a good day. He gets paid per order, so no orders means no money.

A Tuesday in July, without JioMausam

At 2 pm the sky over Andheri turns black. In half an hour a cloudburst floods the roads. Restaurants stop taking orders, and riding a scooter through knee-deep water isn't safe. Ravi waits under a flyover for three hours. He makes ₹180 that day instead of ₹900.

Two kilometres away in Jogeshwari it barely drizzled. The weather report on TV just said "heavy rain in Mumbai". Nobody measured what happened on Ravi's streets, so nobody could help him even if they wanted to. No insurance company sells cover for "I lost ₹700 today", because checking every such claim would cost more than paying it.

The same Tuesday, with JioMausam

Last month, while recharging his phone, Ravi tapped "Add Mausam Kavach for ₹29". That's all he did.

On Tuesday, the towers around Andheri notice their signals getting much weaker: heavy rain. The system works out that 14 mm of rain fell on Ravi's area between 2 and 5 pm. That is more than the 10 mm trigger in his policy. At 5:10 pm his phone buzzes:

MESSAGE ON RAVI'S PHONE

₹300 credited to your JioPay wallet. Mausam Kavach: 14 mm of rain was measured near you between 2:00 and 5:00 pm today. Nothing to do. Stay safe.

He didn't file anything. Nobody came to inspect. It doesn't replace his whole day's earnings, but it covers his fuel and dinner, and it arrived the same evening.

And in May, Sunita

Sunita sells cut fruit from a cart in Ahmedabad. In a heatwave, nobody shops between noon and 4 pm, and standing in 46°C heat is dangerous. With Kavach, when the measured heat in her area crosses the local danger level, she gets paid in the same automatic way.

IN PLAIN WORDS

That's the whole product. Everything else in this guide explains how the towers "know" it rained, how we proved it works, and how it makes money for Reliance.

Why India can't see its own weather

Rain doesn't fall evenly: one neighbourhood floods while the next stays dry. To help people like Ravi, you need to know what happened on *each street*. India measures weather far more coarsely than that.

The three tools India uses today

TOOL	HOW IT WORKS, SIMPLY	ITS WEAKNESS
Weather radar	A big rotating dish on a tower. It sends out radio pulses that bounce off raindrops, like a bat using echoes, and works out where it's raining.	Very expensive (crores each). Only about 40 for the whole country.
Rain gauge	Basically a measuring cup left outside. Whatever water collects is the rainfall.	Measures only the exact spot it's in. Many are read once a day, not live.
Satellite	A camera in space looking down at clouds.	Sees the tops of clouds, not what actually reaches the street. Each picture covers big squares, around 10 km wide.

THINK OF IT LIKE THIS

Imagine trying to watch every street in India with only 40 CCTV cameras. You'd see the big picture, but you'd have no idea what happened on any particular lane. That's India's weather data today.

Why insurance can't help Ravi today

Insurance is simple at heart: many people each pay a small amount into a shared pot, and the few who suffer a loss get paid from it. Normal insurance asks you to **prove your loss**: you file a claim, someone checks it, there's paperwork, and you wait weeks or months.

For a car accident worth ₹50,000, that checking is worth doing. For Ravi's ₹700 lost afternoon, the checking would cost more than the payout. So nobody offers it, and millions of workers go uncovered.

~90%

of Indian workers are in the informal sector: no salary, no paid leave, no cover when weather stops work.

34 million

full-time jobs' worth of working hours India could lose to heat by 2030 (International Labour Organization).

THE PROBLEM IN ONE LINE

Nobody has a cheap, live, street-by-street record of the weather. So nobody can protect workers against it quickly.

Towers that can feel the rain

First: how mobile towers talk to each other

When you make a call, your phone talks to the nearest tower. That tower then has to pass your call on towards the rest of the network. It does this in one of two ways:

- **Fibre cable:** a thin glass cable, usually underground, that carries information as flashes of light. Rain has no effect on it.
- **Microwave link:** where there's no cable, two dishes on two towers point at each other and send a narrow, invisible radio beam across the gap, often 1 to 30 km. The word "microwave" just means the type of radio wave (the same family your microwave oven uses, but far weaker and harmless).

Clear day



Signal arrives strong

Rainy day



Raindrops soak up the beam, so it arrives weaker

What rain does to the beam

Raindrops soak up and scatter part of the beam on its way across. **More rain means a weaker signal at the other end.** The amount of weakening follows a well-known scientific formula, so you can run it backwards: from "how much weaker did the signal get?" to "how hard was it raining?"

THINK OF IT LIKE THIS

Shine a torch at a friend across the road. On a clear night they see it brightly. In heavy rain or fog, the beam looks dimmer by the time it reaches them. If you knew exactly how dim it got, you could tell how heavy the rain was.

The best part: the data already exists

To keep the network healthy, every link already records how strong its signal was when it was **sent** and when it was **received**, every few minutes, like a diary. Engineers use it to spot faults. We use the same diary to read the weather. **No new machines, no new towers.**

SAY IT LIKE THIS

"We don't build sensors. We become the sensor."

Phones as thermometers

Rain is half the story. For heat, we use the phones themselves. Every smartphone has a temperature sensor inside its battery (it's there to stop the battery overheating). A phone's battery warms up and cools down with the air around it.

One phone on its own is unreliable: it might be in a pocket, charging, or in an air-conditioned office. But **average thousands of phones in the same area** and the odd ones cancel out. What's left closely follows the real air temperature. Scientists showed this with real data from eight cities in 2013.

THINK OF IT LIKE THIS

Ask one person to guess how many sweets are in a jar and they could be way off. Ask a thousand people and average their guesses, and the answer is usually surprisingly close. This is called the "wisdom of the crowd".

BE HONEST ABOUT THIS

Our prototype proves the **rain** part on real data. Heat is our second step. The science for it is published, but we haven't built that module yet. If asked, say exactly that.

Is this real, or a crazy idea?

It's real. Scientists have been doing this in research for almost 20 years. What nobody has done is turn it into a product at India's scale.

WHEN	WHO	WHAT HAPPENED
2006	Scientists in Israel	First proved that mobile-network links can measure rain. Published in <i>Science</i> , one of the world's top journals.
2013	Scientists in the Netherlands	Made rain maps of their whole country using one phone company's links.
2013	Same Dutch group	Showed that phone battery temperatures can track city temperatures.
2016 on	Tomorrow.io, a US company	Started as a weather business using wireless signals as sensors.
2020 on	Germany and Sweden, with Ericsson	Thousands of links processed live; data and free software shared with the world.

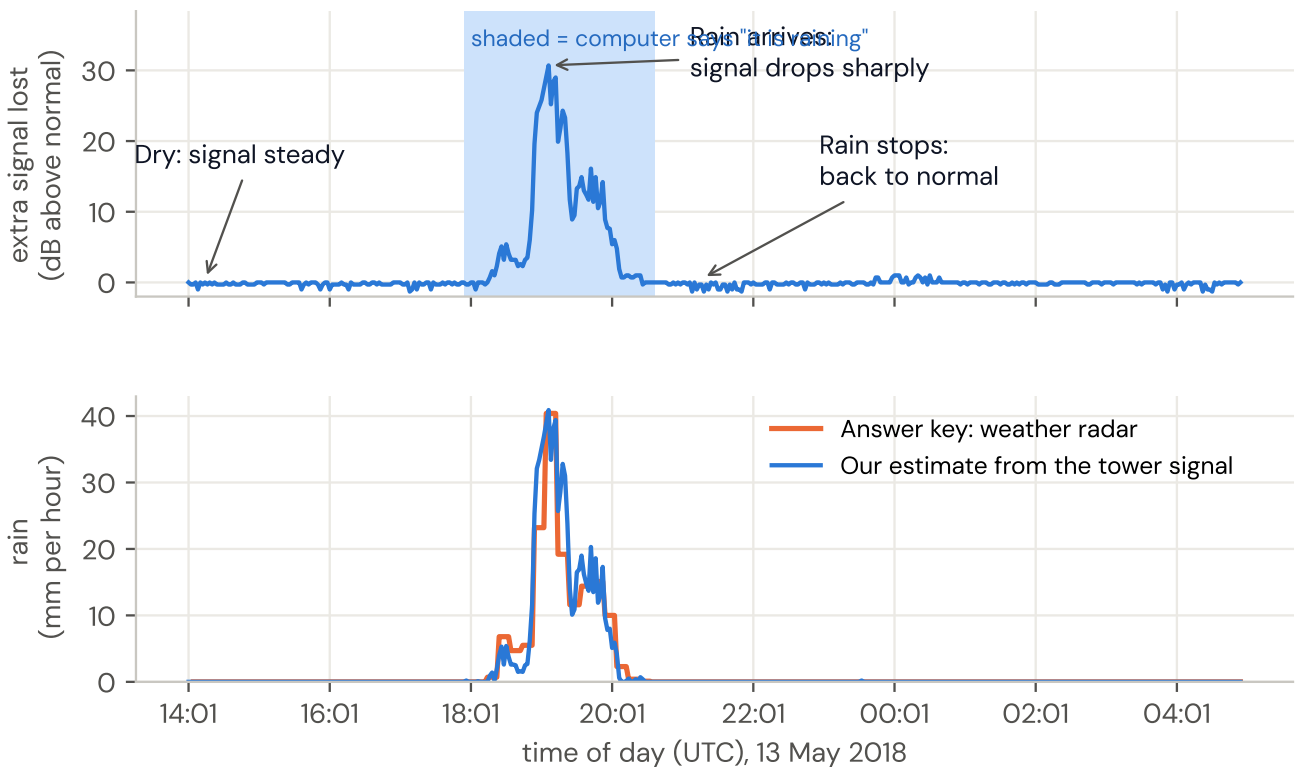
IN PLAIN WORDS

"Published in a journal" or "peer-reviewed" means other independent scientists checked the work before it was printed. It's the gold standard of "this has been verified".

How signal becomes rain, step by step

A computer program does this automatically for every link, every minute. In tech-speak this chain of steps is called a **pipeline**: data goes in one end, and rain comes out the other.

- 1 **Measure the loss.** Take “signal strength sent” minus “signal strength received”. That’s how much signal got lost on the way. It’s measured in **decibels (dB)**, a unit for signal strength. Every 3 dB lost means roughly half the signal is gone.
- 2 **Spot when it’s raining.** On a dry day the loss is steady. When rain arrives it starts to wobble and jump. The computer watches how much it has wobbled over the last hour. If the wobble is unusually large for that particular link, it says “wet”.
- 3 **Know what “normal” looks like.** Every link loses some signal even on a dry day. That normal level is called the **baseline**. Rain is only the extra loss above it. *Like knowing your normal body temperature is 37°C, so you can tell when you have a fever.*
- 4 **Remove the wet-glasses effect.** When it rains, a thin film of water forms on the dish itself and blocks a bit of signal, even in a light drizzle. This is the **wet antenna effect**. We subtract it so we don’t overcount. *Like wet spectacles making everything look blurrier than it really is.*
- 5 **Convert to rain.** A standard formula from the **ITU** (the United Nations body for telecom) turns “extra loss per kilometre” into “millimetres of rain per hour”. *Like a conversion table between inches and centimetres.*



A real example from our prototype: one 7.4 km link during a storm. Top: the signal loss jumps when rain arrives and goes back to normal when it stops. Bottom: the rain we calculated (blue) closely follows the official weather radar (orange).

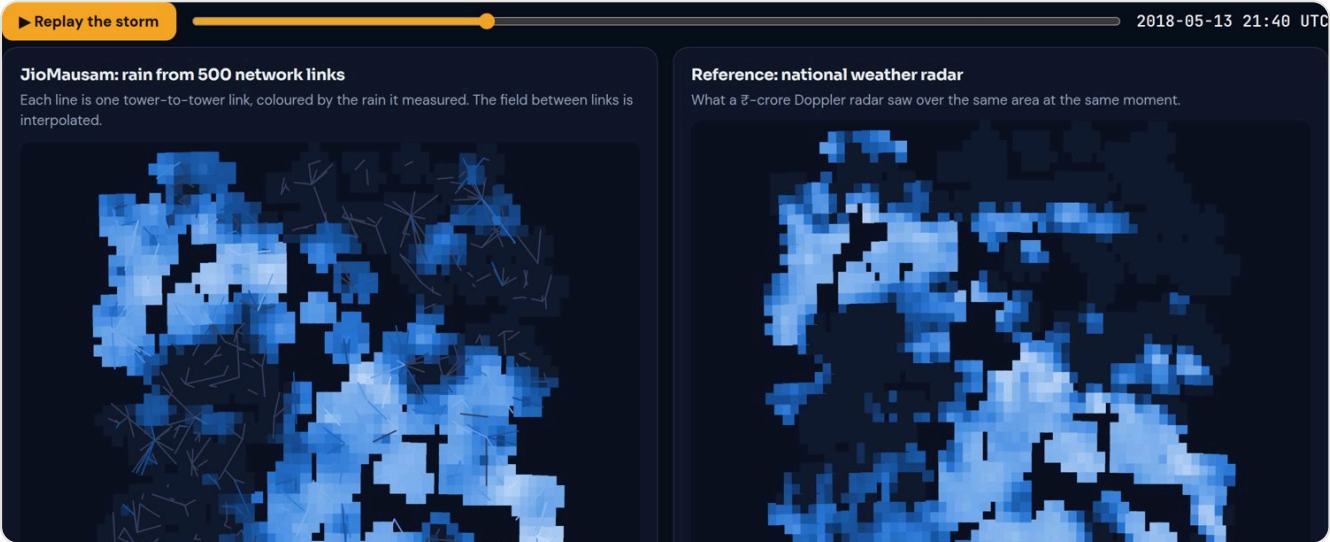
Turning hundreds of links into one map

Each link tells us how hard it rained **along its own line** between two towers. A city has hundreds of these lines criss-crossing it, which gives a web of rain readings.

To draw a map, we split the area into small squares (about 1 km in a real launch, 4 km in our prototype) and estimate each square’s rain from the nearest links. This “filling in the gaps” is called **interpolation**. Our method, **inverse distance weighting**, just means “nearer links count more”. The map refreshes every 10 minutes.

THINK OF IT LIKE THIS

If it’s 30°C in Pune and 34°C in a town 100 km away, your best guess for a village halfway between is about 32°C. If the village is much closer to Pune, you guess nearer 30°C. That’s interpolation.



From our prototype: the same moment of a real storm. Left, our map made only from network links (each thin line is one link). Right, the official weather radar. The shapes and the heavy patches match.

How much rain is “a lot”?

Rain is measured in **millimetres (mm)**: if 1 mm of rain fell, it would make a layer of water 1 mm deep on flat ground. “mm per hour” tells you how fast it’s falling.

RAIN RATE	WHAT IT FEELS LIKE
under 2.5 mm/hour	Light rain. You might skip the umbrella.
2.5 to 7.6 mm/hour	Moderate. Puddles form, and two-wheelers slow down.
over 7.6 mm/hour	Heavy. Roads start to flood and deliveries stop.
100 mm/hour or more	What IMD calls a cloudburst. Flash floods.

Our Kavach trigger of **10 mm within 3 hours** means a few hours of steady heavy rain: the kind that actually stops outdoor work.

Insurance, simply

The basic words

Policy	The agreement: what you're covered for and what you'll get.
Premium	What you pay for the cover. For Kavach, ₹29 a month.
Payout	The money you receive when the covered event happens. For Kavach, ₹300.
Claim	In normal insurance, your request to be paid, with proof. Kavach doesn't need one.
Insurer / underwriter	The licensed company that takes the risk and pays out. Not Jio: a partner insurance company.

Normal insurance vs parametric insurance

Normal insurance ("indemnity")



Parametric insurance (Mausam Kavach)

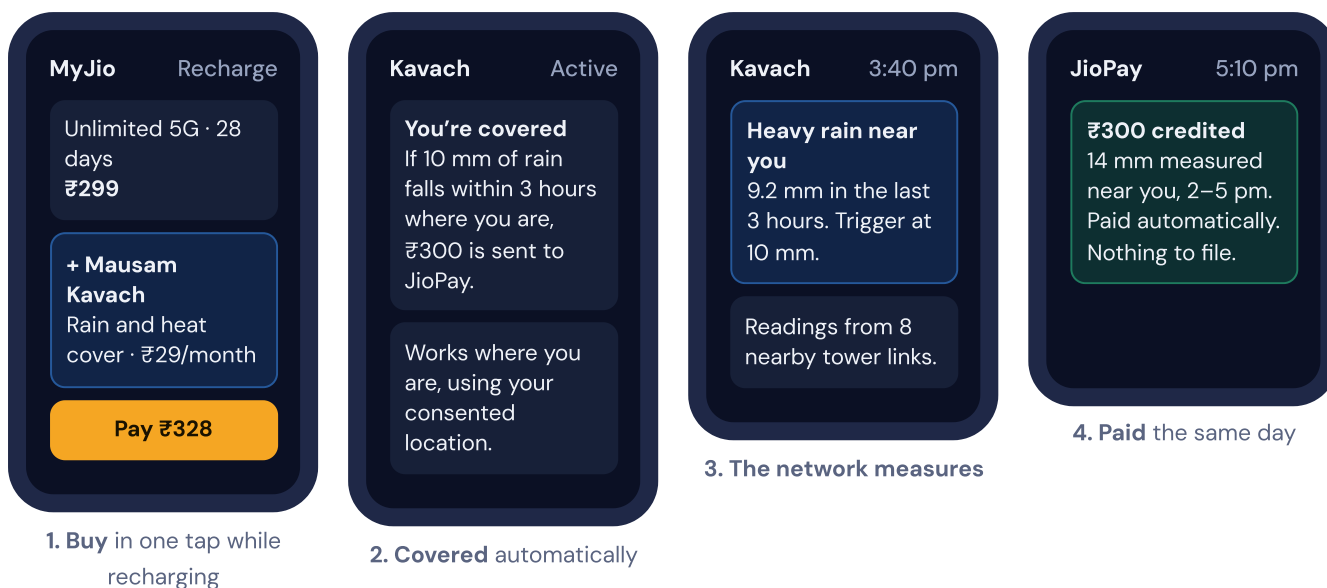


Parametric insurance doesn't pay for your *proven loss*. It pays when a **measured event** happens, according to a rule agreed in advance. That rule is called the **trigger**. For Kavach the rain trigger is "10 mm or more within 3 hours in your 1 km square". Since nobody has to check anything, a tiny ₹300 payout becomes affordable to process.

THINK OF IT LIKE THIS

Some travel insurance pays you automatically if your flight is more than 2 hours late. The airline's official records prove the delay, so you don't send any documents. Kavach works the same way, except the official record is the weather our network measured.

How a payout works, from the customer's side



Why ₹29 a month and ₹300 a payout? The simple maths

- 1 A customer pays $₹29 \times 12$ months = **₹348 a year**.
- 2 We set the trigger so that, on average, a location hits it **about 0.7 times a year**, based on past weather for that square. So the expected payout per customer is $0.7 \times ₹300 = \textbf{₹210 a year}$.
- 3 ₹210 out of ₹348 is about 60%. This share of premiums paid back as payouts is called the **loss ratio**. The remaining 40% covers the insurer's costs and profit, plus Jio's fees.

Triggers are set **for each square separately**, so someone in rainy Mumbai and someone in dry Jaipur pay the same ₹29, but each is protected against what's extreme *for their area*.

Three more words you'll hear

Basis risk	The chance that the rule and real life don't match. Example: Ravi's square got 9 mm (no payout), but his street flooded badly anyway. Or the reverse. We reduce it with small squares, many data sources and a review option for disputes.
Waiting period	Cover starts 7 days after you buy, so nobody can buy it only when a storm is forecast.
Cap	A maximum number of payout days per year, so costs stay predictable.

What we built, and what we tested

A **prototype** is a working early version that shows an idea really works. It isn't the finished product. Think of a model car that actually drives, before the factory is built.

Ours is a website called the **JioMausam Control Room**. It doesn't use made-up numbers: it runs our rain method on **real signal data from 500 microwave links** of a real mobile network, recorded every minute for 11 days in May 2018.

Why the data is from Germany, not India

Phone companies don't publish their link data. Scientists have released one real dataset for research, and it happens to be from Germany (the exact tower locations are shifted for privacy). The physics of rain and radio waves is the same everywhere, and Indian monsoon rain is heavier, which makes the signal even easier to read. Testing on Indian links is exactly what the pilot is for.

How we checked if it works

The same dataset includes readings from Germany's official **weather radar**. We used the radar as our **answer key**: for every link and every hour, we compared "what we calculated" with "what the radar saw".

CALIBRATION

We used the first 3 days to adjust our numbers once, by a single factor ($\times 1.17$), because our method read slightly low. *Like adjusting a kitchen scale that reads a little light.*

HELD-OUT DAYS

We then tested on the other 8 days, which the method had never seen. *Like testing a student on exam questions they haven't practised, so you know they really understand.*

BE HONEST ABOUT THIS

The radar is not perfect either. It's the best "answer key" available, but it has its own errors. When we "disagree" with it, sometimes the radar is the one that's wrong.

The prototype has four screens ("tabs")

Live rain map	Replays a real storm, every 10 minutes: our map next to the radar.
How one link sees rain	The five-step method on one real link (the chart on page 7).
Accuracy	The report card: how close we were to the radar.
Kavach payouts	The insurance in action: 1,000 pretend customers, real storm data, automatic payouts.

Reading our results, one number at a time

First, what “correlation” means

Correlation (written r) is a score from 0 to 1 for how closely two sets of readings rise and fall together. 0 means no relationship at all. 1 means they move perfectly in step.

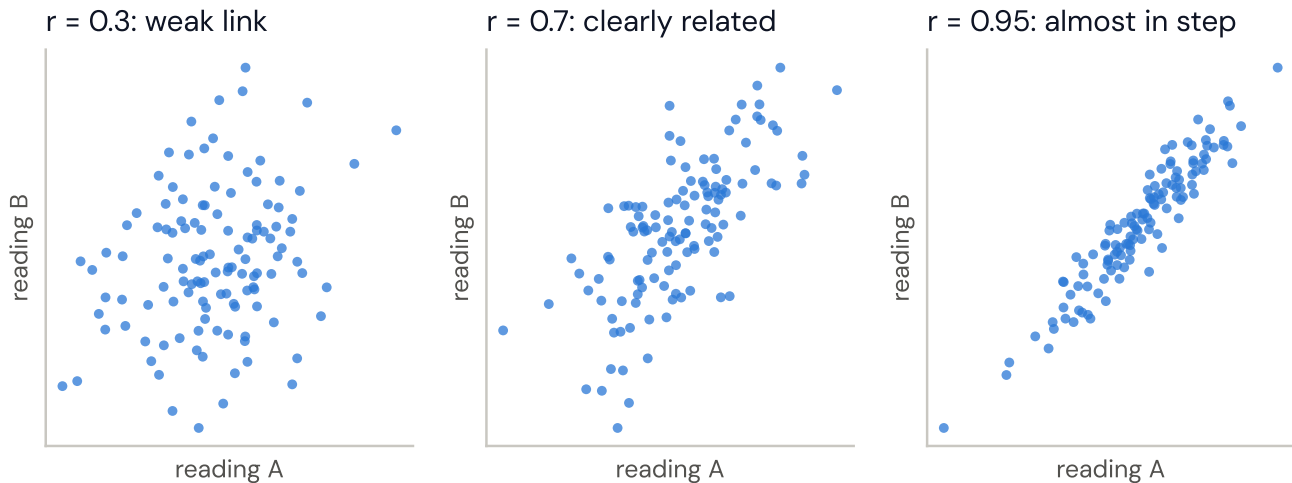


Illustration with made-up dots, just to show what different scores look like. Each dot is one moment: reading A across, reading B up.

0.95

Correlation with radar for the whole network’s average rain, hour by hour.

When the radar said “more rain”, we said “more rain” too, almost perfectly. Say: “Network-wide, we match radar at 0.95.”

0.88

Correlation for a typical single link. Even one pair of towers on its own tracks the radar very well. 91% of links scored above 0.7.

91%

Rainy days caught (the detection rate): of all the days the radar said a link’s path got rain, we said so too 91% of the time. **4.6% false alarms**: of the days we said “rain”, only 4.6% the radar didn’t agree. Like a smoke alarm that rings for 91 of 100 real fires and almost never rings for burnt toast.

94.6%

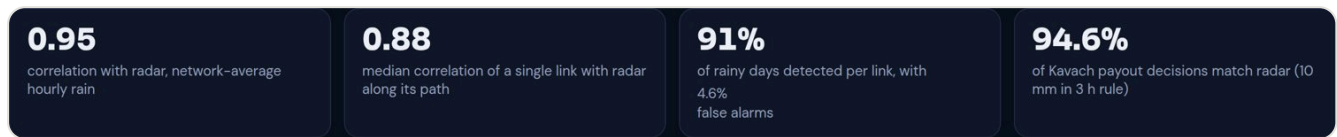
Same payout decision as radar. For every 4 km square on every test day, we asked “would Kavach pay?” using our data and using radar data. They gave the same answer 94.6% of the time.

BE HONEST ABOUT THIS

94.6% sounds great, but it’s high partly because most days are dry, and saying “no payout” on a dry day is easy. (A forecaster in a desert who always says “no rain” is right most days.) The tougher number: of the heavy-rain events the radar saw, we caught **68%**. Also, our rain totals came out about **18% lower** than radar. Both get better in the pilot by adding more links, rain gauges and radar data. Judges respect you more for saying this yourself.

A tour of the Control Room, part 1

Open the link to the prototype in a browser on a laptop. At the top you'll always see four headline numbers. Under them are the four tabs.



The four headline numbers. They're explained on page 12.

Tab 1 • Live rain map



- **What you see:** two maps of the same area at the same moment. Left is ours, right is the radar. Darker blue is lighter rain; brighter blue is heavier.
- **What to click:** "Replay the storm". The real storm of 13–14 May 2018 plays in 10-minute steps. You can drag the slider to any moment.
- **What to say:** "Left is our network, right is a radar worth crores. Same storm, and we didn't need a radar."
- **The boxes underneath:** average rain right now (ours vs radar), how many of the 500 links are currently sensing rain, and how much of the area is getting rain.

A tour of the Control Room, part 2

Tab 2 - How one link sees rain



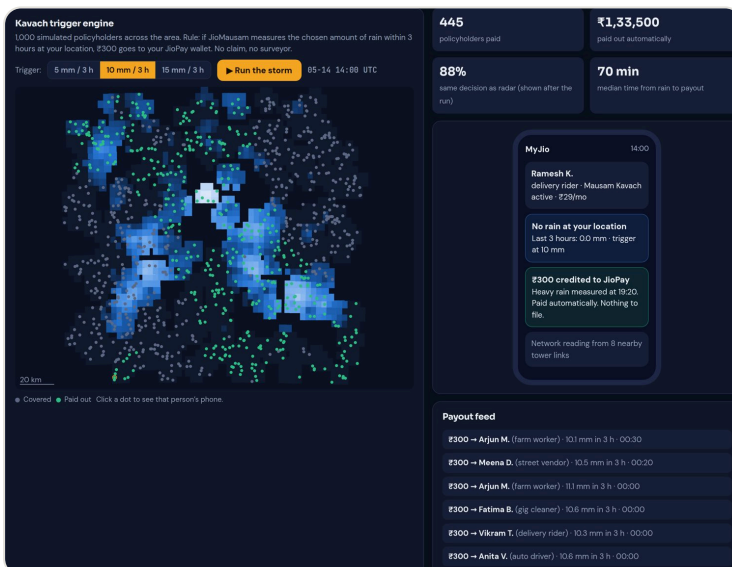
The five steps from page 7 shown on one real link. The top chart is signal loss (blue shaded bands are where the computer decided “raining”). The bottom chart is our rain vs the radar. Hover over a chart to read exact values.

Tab 3 - Accuracy



The report card. Top left: 11 days of average rain, ours and radar’s (grey band = calibration days). Top right: one dot per link per hour; dots near the diagonal line mean we agreed. Bottom: how much of the radar’s rain we caught, with false alarms.

Tab 4 - Kavach payouts



- **The dots** are 1,000 pretend customers spread across the area. Grey = covered, green = paid.
- **The trigger buttons** (5, 10 or 15 mm in 3 hours) change the rule. A lower rule means more payouts, and more chance of paying when it wasn’t needed.
- **“Run the storm”** plays the storm. Watch dots turn green and the payout feed fill up.
- **Click any dot** to see that person’s phone, as in Ravi’s story.
- **The result:** with the 10 mm rule, 445 of 1,000 people were paid ₹1,33,500 in total, about 70 minutes after their rain started.

How the money works

A business idea needs to explain who pays, how much, and why it's cheap to deliver. JioMausam earns in three ways.

WAY TO EARN	WHO PAYS	BUSINESS SHORTHAND
1. Selling Kavach	Workers buy it at recharge. Or companies (like delivery apps) buy it for their workers.	B2C = business to consumer. B2B2C = a company buys it for its people.
2. Selling the weather data	Insurance companies, state disaster departments, city flood teams, farm and logistics companies pay to use our live map.	B2B = business to business. B2G = to government. API = how their computers plug into our data.
3. Reliance uses it itself	JioMart stocks better for rainy days, Reliance New Energy predicts solar power better, Jio predicts weather problems on its own network.	Customer zero = the company is its own first customer.

The famous ₹174 crore, step by step

- 1 Jio has about **500 million** (50 crore) users.
- 2 Suppose just **1%** buy Kavach. That's **5 million** (50 lakh) people.
- 3 Each pays ₹29 × 12 = **₹348 a year**.
- 4 50 lakh × ₹348 = **₹174 crore a year** in premiums. About 60% (₹104 crore) goes back to people as payouts. The rest pays the insurer, Jio's fees and running costs.

IN PLAIN WORDS: WHY THIS WORKS WHEN TINY INSURANCE USUALLY FAILS

Small insurance usually fails for two reasons: finding customers is expensive (agents, ads) and checking claims is expensive. Here, customers are found inside a recharge they already do every month, which costs almost nothing. In business terms, the **customer acquisition cost (CAC)** is near zero. And payouts are automatic, so there are no claims to check.

BE HONEST ABOUT THIS

These are illustrative numbers to show the size of the opportunity, not a forecast. Jio doesn't take the insurance risk itself: a partner insurer does, and Jio earns a **distribution fee** (for selling it) and a **data licence fee** (for providing the trigger data).

Five puzzle pieces that only Reliance holds

To make JioMausam work, you need five things at the same time. Most companies have one or two. Reliance has all five under one roof.

PIECE	WHAT'S NEEDED	RELIANCE ALREADY HAS
Sense	Towers everywhere people live, to measure the weather	The Jio network: lakhs of sites, every district
Reach	A way to reach hundreds of millions of people every month	500M+ Jio users who recharge, and the MyJio app
Pay	A way to send ₹300 instantly and cheaply	JioPay and JioFinance
Insure	A licensed insurance partner	Jio Financial Services and its insurance partners
Use	Businesses that benefit from weather data	JioMart, Reliance Retail, Reliance New Energy, Jio itself

Two business words for this

MOAT

Like the water ditch around a castle: something that makes it hard for competitors to copy you. Our moat is that nobody else holds all five pieces.

FLYWHEEL

A cycle that speeds itself up. More users means more phones and sites, so better maps, so fairer payouts, so more users.

Who else is out there?

- **IMD (India Meteorological Department):** the government weather office. It has radars and forecasts, but not street-level detail, and it doesn't sell insurance.
- **Private weather companies:** sell forecasts, but don't own a sensor network in every Indian street.
- **Insurance companies:** some test weather-triggered products, but lack street-level trigger data and cheap ways to reach workers.
- **Other telecom companies (Airtel, Vi):** could sense rain too, but don't have payments, insurance, retail and energy all together.

The weak spots, and our answers

Every idea has weak spots. Judges will look for them. Knowing them before they ask makes you look prepared, not weak.

THE WORRY	OUR ANSWER, IN SIMPLE WORDS
“Jio mostly uses fibre cables, not microwave beams.” Fibre isn’t affected by rain, so it gives no rain signal.	True for much of the network. But many towers, especially rural, hilly and newer ones, still use microwave links. We also add other signals (phone and tower data) and official radar and gauges. The first job of the pilot is to measure exactly how much coverage we get.
“What if someone loses money but doesn’t get paid?” This is basis risk (page 10).	Small squares, several data sources, rules set for each area, and a review option for complaints. Even before adding other sources, our prototype agreed with radar on 94.6% of payout decisions.
“Isn’t using phone data a privacy problem?”	The rain part uses tower equipment data, not personal data. The heat part only uses phones whose owners agree, combined by area so no single person is visible. This follows India’s data protection law, the DPDP Act 2023 .
“Who allows you to sell insurance?”	The insurance regulator, IRDAI . It runs a regulatory sandbox : a safe trial zone where new products are tested with a limited number of customers. We start there with a licensed partner.
“Will poor workers trust it?”	Delivery apps can buy it for their riders, so it’s free to them. Messages go out in local languages on WhatsApp. And the first automatic payout is the best advertisement.
“Your rain numbers read low.”	About 18% low in the prototype. It’s a known effect of the simple method, fixed by regularly comparing with rain gauges (calibration).
“People will buy it only when a storm is coming.”	A 7-day waiting period, and monthly auto-renewal with the recharge.

IF YOU DON’T KNOW AN ANSWER, SAY THIS

“We haven’t measured that yet. It’s exactly what our 90-day pilot is designed to find out, and here’s what we’d measure.”

What happens next if Reliance says yes

A **pilot** is a small, real-world trial before going big, like opening one shop before opening a hundred. A **roadmap** is the plan of steps and timings.

WHEN	STEP	WHAT HAPPENS, SIMPLY
First 3 months	Pilot in two places	Vidarbha (Maharashtra farm region hit by erratic monsoon) for rain, and Ahmedabad (a city with a heat action plan) for heat. Read Jio's real link data there and check our maps against IMD's instruments.
Months 4–9	Insurance trial	Launch Kavach in IRDAI's sandbox for about 50,000 people, through one delivery app and the MyJio app.
Months 9–18	Ten cities	Start selling weather data to companies and governments. JioMart and New Energy start using it.
Year 2+	All of India	A national live map. Offer the method to telecom companies in other monsoon countries.

How we'll know the pilot worked

- Our rain readings match IMD's at a correlation of **0.8 or more** (the prototype already reached 0.88 per link).
- We catch **at least 80%** of heavy-rain events, with fewer than 15% false triggers.
- Money reaches people's wallets within **6 hours** of the trigger.

What we'd ask Reliance for

- Access to the link data in the pilot areas (handled by Jio's own team).
- A partner in Jio Financial Services and an insurance company.
- A spot in the MyJio app, and an introduction to one delivery company.
- A small team of about five people. **No new hardware.**

IN PLAIN WORDS

The investment is mostly people and computers, not machines. That's a big part of the pitch: Reliance already paid for the expensive part when it built the network.

Every technical word, in plain language

WEATHER AND SCIENCE

Attenuation · The weakening of a signal on its way. Rain causes attenuation. “Signal loss” means the same.

Cloudburst · Extremely intense rain over a small area, about 100 mm or more in an hour (IMD’s definition).

Doppler weather radar · A big rotating dish that bounces radio pulses off raindrops to see rain and its movement. India has about 40.

Heat index · How hot it actually feels, combining temperature and humidity. 40°C in humid Mumbai can feel worse than in dry Jaipur.

Heatwave · A spell of abnormally high temperatures. IMD declares it based on local normal temperatures.

Hyperlocal · At a very small scale: street or neighbourhood, not city or state.

IMD · India Meteorological Department, the government’s weather office.

Mission Mausam · A government programme launched in 2024 (about ₹2,000 crore) to improve India’s weather observation and forecasting.

mm of rain · Depth of rainwater on flat ground. “mm per hour” is how fast it’s falling.

Peer-reviewed · Checked by independent scientists before publication. The gold standard of trust in research.

Rain gauge · A container that collects rain to measure it at one spot.

Satellite data · Pictures and measurements from space. Good for big patterns, not street detail.

Urban heat island · Cities, full of concrete, get hotter than leafy or rural areas nearby.

MOBILE NETWORKS

Antenna / dish · The device on a tower that sends and receives radio signals.

Backhaul · The connections that carry traffic from a tower back into the main network, by fibre cable or microwave link.

Base station / cell site · A mobile tower and its equipment.

CML (commercial microwave link) · The scientists’ name for a microwave link run by a phone company. Our “rain gauges”.

Decibel (dB) · A unit for signal strength. Losing 3 dB means roughly half the signal is gone; 10 dB means 90% gone.

E-band · Very high-frequency microwave links (70–80 GHz), used for fast connections. Very sensitive to rain, which is good for us.

Fibre (optical fibre) · Thin glass cable carrying data as light. Unaffected by rain.

Frequency / GHz · How fast a radio wave vibrates. GHz means billions of times per second. Our links run from 6 to 39 GHz. Higher frequencies are more affected by rain.

Microwave link · A narrow radio beam between two dishes on two towers, often 1–30 km apart.

RSL / TSL · Received Signal Level and Transmitted Signal Level: how strong the signal was when it arrived and when it was sent. The difference is the loss.

Rain fade · When rain weakens a link so much that the connection degrades. A problem for Jio, and a signal for us.

Performance logs · The network’s automatic diary of how each piece of equipment is doing. Where our raw data comes from.

4G / 5G · Generations of mobile technology. 5G is the newest in India.

ISAC · “Integrated sensing and communication”: the research idea, expected in 6G, that networks should also sense their surroundings. JioMausam is an early, practical version of it.

Data, numbers and insurance words

DATA AND NUMBERS

Algorithm · A set of step-by-step instructions a computer follows.

Answer key / reference · The trusted measurement we compare against. For us, weather radar.

Baseline · The normal dry-weather level of a link's signal loss. Rain is measured above it.

Bias · A consistent lean in one direction. Our totals read about 18% low, so we have a low bias.

Calibration · Adjusting a measurement so it matches a trusted reference, like setting a kitchen scale to zero.

Correlation (r) · A 0-to-1 score of how closely two readings rise and fall together. 0.95 means almost in step.

Dashboard · A screen of live charts and numbers. Our Control Room is one.

Data pipeline · The chain of automatic steps that turns raw data into results.

Dataset · A collection of data. Ours: 500 links, 11 days, readings every minute.

Detection rate · Of the real events, the share we caught. Also called "hit rate" or POD (probability of detection).

False alarm rate · Of the alarms we raised, the share that were wrong.

Grid / cell · Splitting a map into equal squares. Each square is a cell.

Held-out data · Data kept aside and used only for the final test, never for tuning.

Interpolation · Estimating values between known points. How we fill the map between links.

Inverse distance weighting (IDW) · An interpolation method where nearer points count more.

Open-source · Software whose code is free for anyone to use and check. We used an open-source tool called pycomlink.

Prototype · A working early version that proves an idea, not the finished product.

Real-time · Updated live, as things happen. Our map refreshes every 10 minutes.

Resolution · How fine the detail is. "1 km, 10 minutes" means 1 km squares updated every 10 minutes.

Standard deviation / "wobble" · How much readings jump around. We use it to spot when rain starts.

Wet antenna effect · Extra loss from a film of water on the dish. We subtract it.

INSURANCE AND MONEY

Actuary · A maths expert who prices insurance.

Basis risk · The chance the trigger and the real loss don't match.

Cap · The maximum number of payouts allowed per year.

Claim · A request for payment with proof, in normal insurance. Kavach needs none.

Indemnity insurance · Normal insurance that pays your proven loss after checking.

Insurer / underwriter · The licensed company that carries the risk and pays out.

IRDAI · Insurance Regulatory and Development Authority of India, the regulator.

Loss ratio · Payouts as a share of premiums. We plan for about 60%.

Micro-insurance · Insurance with very small premiums and payouts, built for low-income people.

Parametric insurance · Pays automatically when a measured index (like rainfall) crosses a pre-agreed trigger.

Payout · The money paid when the trigger fires: ₹300 for Kavach.

PMFBY · Pradhan Mantri Fasal Bima Yojana, the government's crop insurance scheme.

Policy · The insurance agreement.

Premium · What you pay for cover: ₹29 a month for Kavach.

Regulatory sandbox · A trial zone run by a regulator where new products can be tested with limited customers.

Reinsurance · Insurance for insurance companies, so one huge storm can't bankrupt them.

Trigger · The pre-agreed rule that causes a payout, e.g. 10 mm of rain within 3 hours.

Waiting period · Days after buying before cover starts (7 for Kavach), to stop last-minute buying.

Business, strategy and policy words

BUSINESS AND STRATEGY

API · Application Programming Interface: a way for another company's computer system to plug into our data automatically.

B2C / B2B / B2G / B2B2C · Who you sell to: consumers, businesses, government, or a business that buys for its own people.

Business case · The argument for why an idea will make (or save) enough money to be worth doing.

CAC (customer acquisition cost) · What it costs to win one customer. Near zero for Kavach, thanks to the recharge.

Customer zero · When a company is the first user of its own product. Reliance's own businesses use JioMausam first.

Distribution · How a product reaches customers. Ours rides on Jio's recharge.

Distribution fee · What Jio earns for selling the insurance.

Data licence fee · What Jio earns for providing the trigger data.

Flywheel · A self-reinforcing cycle: more users make the product better, which brings more users.

Go-to-market (GTM) · The plan for launching and selling a product.

Gig worker · Someone paid per task through an app: delivery riders, cab drivers. NITI Aayog expects 23.5 million by 2029–30.

Informal sector · Work without formal contracts, salary slips or benefits. About 90% of India's workers.

Moat · What makes a business hard to copy.

Monetise · To turn something into money.

Pilot · A small real-world trial before full launch.

Platform · A base system other products are built on. JioMausam is the platform; Kavach is one product on it.

Roadmap · The planned steps and timeline.

Scalable · Can grow to many more users without costs growing as fast.

Unit economics · The money made or lost on one customer: here ₹348 premium, about ₹210 expected payout.

Value proposition · Why a customer should care. For Ravi: "₹29 a month, paid automatically when weather stops your work."

RELIANCE COMPANIES

Jio Platforms · Reliance's digital and telecom arm, including the Jio network and MyJio app.

JioPay / JioFinance · Reliance's payment and finance apps, our payout channel.

Jio Financial Services (JFS) · Reliance's listed financial services company, which works with insurance partners.

JioMart · Reliance's online grocery and quick-delivery business.

Reliance New Energy · Reliance's clean energy business (solar, batteries, hydrogen).

GOVERNMENT AND RULES

DPDP Act 2023 · Digital Personal Data Protection Act: India's law on how personal data is collected and used, with consent.

Consent · The user clearly agreeing to how their data is used.

Anonymised / aggregated data · Data with names removed and combined by area, so no individual can be identified.

State disaster management authority · The state body that handles floods, heatwaves and other emergencies. A likely buyer of our data.

Heat action plan · A city's plan to protect people during heatwaves. Ahmedabad was the first in South Asia.

How to explain it to anyone

10 SECONDS

“Rain weakens the signals between mobile towers. We use that to map weather street by street, and pay workers automatically when it stops their work.”

30 SECONDS

“India has only about 40 weather radars, so nobody knows which streets got hit by a storm. But Jio’s towers talk over radio beams, and rain weakens those beams in a measurable way. JioMausam reads that to build a live rain map with no new hardware. On top, Mausam Kavach lets a delivery rider pay ₹29 a month with their recharge and get ₹300 automatically when heavy rain hits their area. We tested it on 500 real links: it matches official radar at 0.95.”

2 minutes: the structure

1. **Hook:** “India’s biggest weather network already exists. It’s called Jio.”
2. **Problem:** weather is street-level, data isn’t; workers lose income; insurance is too slow.
3. **Insight:** rain weakens tower signals; phones feel heat. We become the sensor.
4. **Proof:** 500 real links, 0.95 match with radar, 91% of rainy days caught.
5. **Product:** JioMausam map + Kavach ₹29/month, ₹300 automatic payouts.
6. **Business:** premiums, data sales, Reliance as customer zero. ₹174 crore at 1% adoption.
7. **Why Reliance:** network, users, payments, insurance and businesses, all under one roof.
8. **Ask:** a 90-day pilot in Vidarbha and Ahmedabad.

Explain it to your grandmother

“You know how the TV signal goes fuzzy when it rains hard? Mobile towers have the same problem. We use that fuzziness to tell exactly where it’s raining. Then, if a poor worker has paid ₹29 a month and it pours where they are, the phone company sends them ₹300 that same day, without any forms.”

If you forget everything else, remember these numbers

~40	weather radars in all of India
500M	Jio users (our reach)
500	real links we tested on
0.95	our match with radar (network-wide)
91%	rainy days caught (4.6% false alarms)
₹29 / ₹300	monthly price / payout
₹174 Cr	yearly premiums if 1% of Jio users buy

Questions you might be nervous about

Short, simple answers. For deeper versions, see the Q&A pages in the main idea brief.

“How do you know it actually works?”

We tested it on 500 real network links and compared with official weather radar. Network-wide our readings matched at 0.95 out of 1, and we caught 91% of rainy days. Scientists have shown the same thing in Israel, the Netherlands, Germany and Sweden.

“Why is your data from Germany?”

Telecom companies don't publish their data. The only open research dataset is German. The physics is the same in India, and our heavier monsoon rain gives an even clearer signal. Indian data is what the pilot is for.

“Doesn't Jio use fibre cables?”

Much of it, yes, and fibre doesn't sense rain. But many sites still use microwave links, and we add phone data plus official radar and gauges. Measuring real coverage is the pilot's first job.

“Is Jio becoming an insurance company?”

No. A licensed insurer carries the risk. Jio sells it at recharge and provides the weather data, earning fees for both.

“Why would a poor worker pay?”

₹29 a month is less than a cup of tea a week, it's added in one tap, and it pays without any forms. Often their delivery app would pay for them.

“What about privacy?”

Rain uses tower equipment data, not personal data. Heat uses only phones whose owners agree, combined by area, following the DPDP Act.

“What if it rains on someone and they don't get paid?”

That's called basis risk. We keep the squares small, combine several data sources, and offer a review option. Our test already agreed with radar on 94.6% of payout decisions, and the pilot will push that higher.

“Can't Airtel copy this?”

They could read rain, but they don't have payments, insurance, retail and energy businesses to use it. And whoever maps India first owns the best data.

“What will it cost Reliance?”

Mostly a small team and computing. No new towers or machines. The network was the expensive part, and it's already built.

“What's the biggest risk?”

Not having enough microwave links in some cities. That's why the pilot measures it first, and why we combine several data sources instead of relying on one.

Ten-question self-test

Cover the answers, answer out loud, then check. If you get 8 or more right, you're ready for the jury.

1. In one sentence, what does JioMausam do?
2. What happens to a microwave link's signal when it rains, and why?
3. What is a "baseline", and why do we need it?
4. What is the difference between normal insurance and parametric insurance?
5. What is Kavach's price, payout and rain trigger?
6. What does a correlation of 0.95 mean?
7. Why did we test on "held-out days"?
8. Why is 94.6% agreement not the whole story, and what's the tougher number?
9. Name the five puzzle pieces only Reliance holds.
10. What's the biggest risk, and how does the pilot handle it?

Answers

1. It uses the weakening of mobile-tower radio signals (and phone temperatures) to map weather street by street, and powers automatic insurance payouts.
2. It gets weaker, because raindrops soak up and scatter the beam. More rain, more weakening.
3. The link's normal dry-day signal loss. Rain is only the extra loss above it, so without the baseline we'd count normal loss as rain.
4. Normal insurance pays a proven loss after checking a claim. Parametric pays automatically when a measured event crosses a pre-agreed trigger.
5. ₹29 a month, ₹300 per trigger, and 10 mm or more of rain within 3 hours in your square.
6. Our readings and the radar's rise and fall almost perfectly together.
7. To prove the method works on data it was never tuned on, like an exam with unseen questions.
8. Most days are dry, so "no payout" is easy to get right. The tougher number: we caught 68% of the radar's heavy-rain events.
9. Sense (network), reach (500M users), pay (JioPay), insure (JFS and partners), use (JioMart, New Energy, Jio).
10. Too few microwave links in some areas (much of the network is fibre). The pilot measures real coverage first, and we combine several data sources.

YOU'RE READY WHEN

You can tell Ravi's story, explain the torch-in-the-rain idea, and say "0.95 against radar on 500 real links" without looking at notes.